Mathematical Background for the Learnable Kóczy Fuzzy Signature Layer

(HSiKAN as differentiable fuzzy-signature hypergraph convolution)

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2026-05-30

Abstract

We assemble the classical fuzzy-logic foundations that justify a particular neural-network architecture — the *learnable Kóczy fuzzy signature layer* — as a parameterised, differentiable instantiation of objects long studied in fuzzy systems theory. The exposition is organised as a vertical stack: from membership functions and triangular norms, through Atanassov intuitionistic fuzzy sets and Kóczy's hierarchical fuzzy signatures, to defuzzification, Zadeh's linguistic hedges, and the design of *learnable gates*. The endpoint is a precise mathematical specification of the proposed layer in which every learnable parameter is in bijection with a named fuzzy-logic primitive's plasticity hook. We pay particular attention to two design points: (i) the **Atanassov** μ^+/μ^- **pair** reading of HSiKAN's signed branches, and (ii) the **learnable dilation parameter** τ that smoothly connects the crisp signum mixer to the linguistic-hedge family of Zadeh.

Contents

1 Introduction and scope

A *fuzzy signature* in the sense of Kóczy and Vámos [5, 6] is a hierarchical, structured representation of a fuzzy concept in which an internal node aggregates the fuzzy values of its children by an aggregation operator (typically a *t-norm*, *t-conorm*, or weighted average), and leaves carry elementary memberships of features. The signature replaces a single linguistic variable by a labelled tree of linguistic variables, each with its own membership function and its own aggregator.

A *hypergraph convolution* of the HSiKAN family [17, 18] operates on an incidence relation $\mathbf{H} \in [0,1]^{V \times E}$ between vertices V and hyperedges E. Each layer performs a vertex–to–edge gather, an edge-side activation, and an edge–to–vertex scatter; the activation per edge is a learnable spline (Catmull–Rom) whose control points are the trained parameters.

The thesis of this background document is structural:

- Every Kóczy-fuzzy-signature primitive has a structural analog inside an HSiKAN-style hypergraph layer; nothing must be *added* to gain a fuzzy interpretation — instead, mechanisms already present must be *identified* with their fuzzy-logic counterparts.
- The HSiKAN "signed branch" construction is exactly the intuitionistic-fuzzy (μ, ν) pair of Atanassov, with the *polarity gate* playing the role of the hedge that mixes the two memberships.
- Replacing the fixed polarity gate temperature (in HSiKAN, the constant 4) with a learnable scalar τ per channel/layer instantiates Zadeh's hedge family in a smooth, gradient-friendly way: $\tau \to \infty$ recovers the crisp signum mixer, $\tau \to 0$ recovers a uniform $\frac{1}{2}$ mix, intermediate τ sweeps the family of dilations and concentrations.

The motivating engineering problem is to design a vision/robotics layer that is (i) data-efficient (few parameters) by absorbing structural priors as non-trainable scaffolding, (ii) interpretable in classical fuzzy-systems language, and (iii) free from generic neural primitives (arbitrary `Linear` layers, ad-hoc sigmoids) that obscure the fuzzy interpretation. We aim for a layer in which every *learnable* number is a named fuzzy-system primitive.

The remainder of this document is organised as follows. Section 2 reviews membership functions, t-norms, t-conorms, and negations. Section 3 reviews Atanassov intuitionistic fuzzy sets and extracts the precise sense in which HSiKAN's signed branches are an IFS representation. Section 4 reviews Kóczy fuzzy signatures and shows that an HSiKAN hypergraph layer is a depth-1 fuzzy signature in disguise. Section 5 reviews the major defuzzification methods and selects the differentiable Sugeno-style weighted-average as the head of choice. Section 6 introduces Zadeh's linguistic hedges and the learnable-gate construction. Section 7 states the HSiKAN–Kóczy correspondence formally and accounts for every parameter. Section 8 specifies the proposed FuzzySignatureLayer as the smallest layer whose every degree of freedom is one of the primitives above, and proves elementary $[0,1]$-boundary preservation lemmas.

2 Classical fuzzy-logic primitives

2.1 Fuzzy sets and membership functions

Definition 2.1 (Fuzzy set, after [1])**.** *A* fuzzy set *on a universe X is a function $\mu : X \to [0,1]$. The value $\mu(x)$ is the* degree of membership *of x in the set. Crisp sets are recovered as the special case $\mu : X \to \{0,1\}$.*

Standard parametric membership-function families on $X \subseteq \mathbb{R}$ are *triangular* ($\mu(x) = \max(0, 1 - |x - c|/w)$), *trapezoidal* (piecewise-linear with two slopes and a plateau), *Gaussian* ($\mu(x) = \exp(-(x-c)^2/(2\sigma^2))$), and *sigmoidal* ($\mu(x) = \sigma(a(x-c))$, where σ is the logistic function). All four are uniformly continuous and lie in $[0,1]$.

For a learning problem the practitioner must either commit to a parametric family (and learn its few parameters) or admit a *shape-free* learnable membership function. The latter is conveniently realised by a low-order spline of m control points; we use the cubic Catmull–Rom spline (Section 7), whose interpolant has C^1 continuity and is computed from a sliding window of four neighbouring control points.

2.2 Triangular norms (fuzzy AND)

Definition 2.2 (T-norm)**.** *A triangular norm (t-norm) is a binary operator* $\mathsf{T} : [0,1]\times[0,1] \to [0,1]$ *that is* commutative, associative, monotonic *($a \le a' \Rightarrow \mathsf{T}(a,b) \le \mathsf{T}(a',b)$), and has* 1 *as the identity element:* $\mathsf{T}(1,a) = a$*. As a consequence,* $\mathsf{T}(0,a) = 0$ *for all* a*.*

By associativity, T extends uniquely to an n-ary operator $\mathsf{T}(a_1,\dots,a_n)$. The boundary identities $\mathsf{T}(1,\dots,1) = 1$ and $\mathsf{T}(0,a_2,\dots,a_n) = 0$ are immediate.

The three *canonical* t-norms generate, by parameterised interpolation, essentially every t-norm of interest:

$$\mathsf{T}_{\mathrm{G}}(a,b) = \min(a,b) \qquad \text{Gödel / Zadeh,} \tag{1}$$
$$\mathsf{T}_{\mathrm{P}}(a,b) = a\cdot b \qquad \text{algebraic / product,} \tag{2}$$
$$\mathsf{T}_{\mathrm{L}}(a,b) = \max(0,a+b-1) \qquad \text{Łukasiewicz.} \tag{3}$$

Proposition 2.1 (Ordering of canonical t-norms)**.** *For all* $a,b \in [0,1]$*,*

$$\mathsf{T}_{\mathrm{L}}(a,b) \;\le\; \mathsf{T}_{\mathrm{P}}(a,b) \;\le\; \mathsf{T}_{\mathrm{G}}(a,b).$$

Proof. The right inequality is $ab \le \min(a,b)$, which holds because $\min(a,b) \le 1$ and the other factor is $\le \min(a,b)$ by case analysis. The left inequality $\max(0,a+b-1) \le ab$ rewrites as $ab - a - b + 1 = (1-a)(1-b) \ge 0$, which is true. □

Parameterised t-norm families. Three families are common and admit a single scalar parameter:

- **Hamacher** [9], parameter $\gamma \ge 0$:

$$\mathsf{T}^{\mathrm{H}}_{\gamma}(a,b) = \frac{ab}{\gamma + (1-\gamma)(a+b-ab)}, \qquad \gamma = 1 \Rightarrow \mathsf{T}_{\mathrm{P}}.$$

- **Yager** [7], parameter $p > 0$:

$$\mathsf{T}^{\mathrm{Y}}_{p}(a,b) = 1 - \min\!\Big(1, \big((1-a)^p + (1-b)^p\big)^{1/p}\Big), \qquad p \to \infty \Rightarrow \mathsf{T}_{\mathrm{G}}.$$

- **Aczél–Alsina** [10], parameter $\lambda > 0$:

$$\mathsf{T}^{\mathrm{AA}}_{\lambda}(a,b) = \exp\!\Big(-\big[(-\log a)^{\lambda} + (-\log b)^{\lambda}\big]^{1/\lambda}\Big), \qquad \lambda = 1 \Rightarrow \mathsf{T}_{\mathrm{P}}.$$

Remark 2.2. *Each parameterised family above is differentiable in its scalar parameter. In a learnable layer, this opens a third axis of plasticity (alongside control points and gate temperature): the t-norm's* family parameter*. The proposed layer (Section 8) holds this lever in reserve and defaults to the three canonical t-norms only, because a single categorical choice empirically suffices for the vision-scale problems in scope.*

2.3 Triangular conorms (fuzzy OR)

Definition 2.3 (T-conorm)**.** *A* triangular conorm *(t-conorm) is a binary operator* $\mathsf{S} : [0,1] \times [0,1] \to [0,1]$*, commutative, associative, monotonic, with* 0 *as identity:* $\mathsf{S}(0,a) = a$*. Consequently* $\mathsf{S}(1,a) = 1$*.*

Each t-norm T has a *dual* t-conorm $\mathsf{S}(a,b) = 1 - \mathsf{T}(1-a, 1-b)$, the De Morgan dual. The three canonical duals are:

$$\mathsf{S}_{\mathrm{G}}(a,b) = \max(a,b) \qquad \text{(Gödel/Zadeh OR)} \tag{4}$$
$$\mathsf{S}_{\mathrm{P}}(a,b) = a + b - ab \qquad \text{(probabilistic sum)} \tag{5}$$
$$\mathsf{S}_{\mathrm{L}}(a,b) = \min(1, a+b) \qquad \text{(Łukasiewicz bounded sum)} \tag{6}$$

Proposition 2.3 (Ordering of canonical t-conorms)**.** *For all* $a, b \in [0,1]$*,*

$$\mathsf{S}_{\mathrm{G}}(a,b) \;\le\; \mathsf{S}_{\mathrm{P}}(a,b) \;\le\; \mathsf{S}_{\mathrm{L}}(a,b),$$

dual to Proposition 2.1.

2.4 Negations

Definition 2.4 (Negation operator)**.** *A* negation *is an order-reversing involution* $\mathsf{N} : [0,1] \to [0,1]$*,* $\mathsf{N}(0) = 1$*,* $\mathsf{N}(1) = 0$*,* $\mathsf{N}(\mathsf{N}(a)) = a$*.*

The standard negation is $\mathsf{N}_s(a) = 1 - a$. Two parametric families with one scalar parameter: *Sugeno* $\mathsf{N}_\lambda(a) = (1-a)/(1+\lambda a)$ $(\lambda > -1)$, and *Yager* $\mathsf{N}_w(a) = (1 - a^w)^{1/w}$ $(w > 0)$.

De Morgan triple. A t-norm, its dual t-conorm, and a negation form a *De Morgan triple* when $\mathsf{S}(a,b) = \mathsf{N}(\mathsf{T}(\mathsf{N}(a), \mathsf{N}(b)))$. The canonical (Łukasiewicz, Łukasiewicz, $1 - \cdot$) is the unique De Morgan triple in which all three operators are continuous and the t-conorm is Archimedean. The proposed layer uses the standard negation only; alternative negations become a future axis.

3 Atanassov intuitionistic fuzzy sets

3.1 The (μ, ν, π) decomposition

Definition 3.1 (Intuitionistic fuzzy set, after Atanassov [4])**.** *An* Atanassov intuitionistic fuzzy set *(IFS) on a universe* X *is a pair of functions* (μ, ν) *with* $\mu, \nu : X \to [0,1]$ *and*

$$\mu(x) + \nu(x) \;\le\; 1 \qquad \textit{for all } x \in X.$$

The slack $\pi(x) = 1 - \mu(x) - \nu(x) \in [0,1]$ *is the* hesitancy index*: the degree of indeterminacy at* x*.*

A classical fuzzy set (Definition 2.1) is recovered as the case $\pi \equiv 0$, i.e. $\nu = 1 - \mu$. The IFS strictly generalises the fuzzy set by admitting a quantified ignorance of the (μ, ν) pair.

3.2 Score and accuracy functions

Definition 3.2 (Score function)**.** *The* score *of an IFS at* x *is* $\mathcal{S}(x) = \mu(x) - \nu(x) \in [-1, 1]$*.*

The score is a signed, scalar summary: $\mathcal{S}(x) > 0$ indicates net membership, $\mathcal{S}(x) < 0$ net non-membership, and $\mathcal{S}(x) = 0$ either balanced support or full hesitancy. In any downstream aggregation, the score plays the role analogous to a real-valued "polarity" signal in a signed network.

3.3 IFS aggregation

Two natural aggregators on a finite family of IFSs $\{(\mu_i, \nu_i)\}_{i=1}^{n}$ are

$$(\mathsf{T}\text{-aggregate}) \quad \big(\mathsf{T}(\mu_1, \dots, \mu_n),\ \mathsf{S}(\nu_1, \dots, \nu_n)\big) \tag{7}$$

$$(\mathsf{S}\text{-aggregate}) \quad \big(\mathsf{S}(\mu_1, \dots, \mu_n),\ \mathsf{T}(\nu_1, \dots, \nu_n)\big) \tag{8}$$

Both preserve the IFS constraint $\mu + \nu \leq 1$. The first models "conjunction of memberships and disjunction of non-memberships" (a strict consensus over the IFSs); the second models the dual.

3.4 The signed-branch reading of HSiKAN

In HSiKAN-vision [17] the per-pixel *polarity* $p_v = \operatorname{sgn}(x_v - \mu_e)$ at a pixel–patch pair (v, e) selects between two Catmull–Rom activations: CR^+ for $p_v > 0$ and CR^- for $p_v < 0$. The smooth gate

$$g(x) = \sigma\big(4 \cdot (x - \mu_e)\big) \in [0, 1], \qquad \operatorname{out}(x) = g(x)\,\mathrm{CR}^+(x) + (1 - g(x))\,\mathrm{CR}^-(x)$$

is a convex combination of CR^+ and CR^- governed by the polarity sign.

If CR^+ and CR^- are constrained to take values in $[0, 1]$ (via the spline clamp $\sigma \circ \mathrm{CR}$, see Section 8), the construction is *exactly* an Atanassov pair with

$$\begin{aligned}
\mu^+(x) &:= \mathrm{CR}^+(x), & \mu^-(x) &:= \mathrm{CR}^-(x),\\
\mu(x) &:= g(x)\,\mu^+(x), & \nu(x) &:= (1 - g(x))\,\mu^-(x),\\
\pi(x) &:= 1 - \mu(x) - \nu(x).
\end{aligned}$$

The constraint $\mu(x) + \nu(x) \leq 1$ follows from the convex combination together with $\mu^\pm \in [0, 1]$, because $g\mu^+ + (1 - g)\mu^- \leq g + (1 - g) = 1$. Thus HSiKAN's signed-branch construction is structurally an IFS with an explicit, learnable (μ^+, μ^-) pair and a polarity-driven gate.

Remark 3.1. *The reference point μ_e (patch mean) in the polarity gate g is a* design choice*, not a fuzzy-logic requirement. A pure IFS reading of HSiKAN drops μ_e and makes the gate*

$$g_\tau(x) = \sigma(\tau\,(x - c))$$

with a fixed centre $c \in [0, 1]$ (e.g. $c = \frac{1}{2}$, the midpoint of the fuzzy range) and learnable dilation τ. This eliminates the non-local dependence of μ_e on the entire patch, simplifies the gradient, and yields the cleanest mapping to the Atanassov framework. The proposed layer in Section 8 adopts this design.

4 Kóczy fuzzy signatures

4.1 Definition

Definition 4.1 (Fuzzy signature, after Kóczy [5])**.** *A* fuzzy signature *is a finite, rooted, labelled tree $\mathcal{T} = (V, E, \ell, A)$ in which*

- *every leaf $v \in V_\ell$ carries a fuzzy value $\mu_v \in [0, 1]$,*
- *every internal node $v \in V_i$ carries an* aggregation operator *A_v (a t-norm, t-conorm, weighted average, OWA, or Sugeno integral) of arity equal to the number of children of v, and*
- *the value at v is recursively $\mu_v = A_v\big(\mu_{c_1}, \dots, \mu_{c_k}\big)$ over its children.*

A fuzzy signature is a structured fuzzy concept: each subtree represents a sub-concept aggregated by its own operator. The root's value is the single fuzzy "score" of the overall concept.

4.2 Hypergraph aggregations as depth-1 signatures

In a hypergraph $H = (V_H, E_H)$ with signed incidence $\mathbf{H} \in [0,1]^{V_H \times E_H}$, the standard "vertex-to-edge" aggregation $\mathbf{h}_e = \mathrm{AGG}\big(\{\mathbf{x}_v : v \in e\}\big)$ is precisely the value of an internal node e of a depth-1 fuzzy signature whose children are the leaves $\{v : v \in e\}$, with $A_e = \mathrm{AGG}$.

A stack of L hypergraph layers, where each layer recomputes $\mathbf{x}_v^{(\ell)} = \mathrm{AGG}'\big(\{\mathbf{h}_e^{(\ell)} : v \in e\}\big)$, is a sequence of depth-1 signatures rather than one deep signature, but *informally* can be read as a fuzzy signature whose depth is L and whose structure is determined by the incidence relation $\mathbf{H}$.

Remark 4.1. *HSiKAN's* multi-arity *construction — running independent incidences* $\mathbf{H}_k$ *for several receptive-field sizes* k *and mixing their outputs by a learnable softmax* α_k *— is, in fuzzy-signature terms, an* OWA across granularities*: the family* $\{\alpha_k\}$ *is a set of OWA weights, normalised by softmax, that aggregate concepts at different scales. Section 6 returns to OWA in the context of learnable gates.*

4.3 The aggregation operator at each node

The choice of A_v at an internal node is the modeller's prerogative. For a learnable signature, two layers of choice apply:

- **Categorical**: a finite menu of aggregators $(\mathsf{T}_\mathrm{G}, \mathsf{T}_\mathrm{P}, \mathsf{T}_\mathrm{L}, \mathsf{S}_\mathrm{G}, \mathsf{S}_\mathrm{P}, \mathsf{S}_\mathrm{L})$. The choice is a hyperparameter set per layer or per node; not learnable by gradient descent (no continuous relaxation in the proposed layer, though see Section 6 for a route).
- **Parametric**: a parameterised family (Hamacher, Yager, Aczél-Alsina) with one learnable scalar per node or per layer. Continuous in the parameter; learnable by gradient descent (Remark 2.2).

The proposed layer uses categorical choice for the t-norm and t-conorm (two hyperparameters) and a parametric learnable scalar τ for the Atanassov gate (Section 6). The categorical aggregation remains a knob at the layer level.

5 Defuzzification

A fuzzy signature, evaluated at its root, returns a fuzzy value $\mu_r \in [0,1]$ (or, for vector-valued signatures, $\mu_r \in [0,1]^d$). For classification or regression a crisp output is required. The *defuzzification* step maps the fuzzy output to a crisp one.

5.1 Methods

Definition 5.1 (Centre of Gravity, COG)**.** *Given a fuzzy output* $\mu : X \to [0,1]$ *on a (typically real) universe* X *and an integrable reference grid,*

$$\mathrm{COG}(\mu) \;=\; \frac{\int_X x\,\mu(x)\,dx}{\int_X \mu(x)\,dx}.$$

Definition 5.2 (Sugeno weighted average, Sug)**.** *Given a finite set of fuzzy values* $\{\mu_i\}_{i=1}^n$ *associated with output levels* $\{y_i\}_{i=1}^n$*,*

$$\mathrm{Sug}(\{\mu_i\};\{y_i\}) \;=\; \frac{\sum_{i=1}^n \mu_i\, y_i}{\sum_{i=1}^n \mu_i}.$$

Definition 5.3 (Mean of Maxima, MoM)**.** $\mathrm{MoM}(\mu) = \mathrm{mean}\{x : \mu(x) = \max_{x'} \mu(x')\}$.

Definition 5.4 (Max-Membership)**.** $\mathrm{MaxMemb}(\mu) = \arg\max_x \mu(x)$.

Definition 5.5 (Bisector)**.** *The bisector splits the area under* μ *into two equal halves:* $\mathrm{Bis}(\mu) = b$ *such that* $\int_{-\infty}^b \mu(x)\,dx = \int_b^{+\infty} \mu(x)\,dx$.

5.2 Differentiability and the choice for a learnable layer

For a layer that participates in a gradient-descent training loop, the defuzzification step must be differentiable in μ almost everywhere. The following table summarises:

Method	Differentiable in μ	Smooth in output position
COG	yes	yes
Sugeno (weighted average)	yes	yes (provided $\sum_i \mu_i > 0$)
MoM	no (subgradient only)	no
Max-Membership	no (argmax)	no
Bisector	yes (implicit)	yes

Of the differentiable options, Sugeno's weighted average has the cleanest analytic form and is the canonical defuzzification for the *Takagi–Sugeno–Kang* (TSK) fuzzy systems most closely related to neuro-fuzzy models. In a vector-valued layer the natural realisation is

$$\mathbf{z} \;=\; \frac{1}{|V|}\sum_{v\in V}\mu_v \;=\; \mathrm{mean}_v(\mu_v) \tag{9}$$

with $\mu_v \in [0,1]^d$ the vector fuzzy output per vertex. The classifier *head*, in the proposed layer, then performs a final linear projection from $\mathbf{z} \in [0,1]^d$ to class logits in $\mathbb{R}^{n_{\mathrm{cls}}}$. This single linear layer is the analytical *Sugeno output parameter*: each row of $W_{\mathrm{head}} \in \mathbb{R}^{n_{\mathrm{cls}}\times d}$ specifies the crisp output value contributed by each fuzzy channel at each class. It is the one place where a generic affine map is unavoidable to map from fuzzy memberships to unconstrained class logits, and it is classically interpreted in Sugeno's TSK framework, not as a generic `Linear` layer.

5.3 Defuzzification in vector form

Sugeno's defuzzification (Equation 9) has two mathematical roles in the proposed layer:

1. **Spatial pool**: average over vertices (pixels), turning a per-pixel fuzzy distribution into a global fuzzy concept (one per channel).
2. **Channel projection**: a linear map (Sugeno output rule) converts the channel-wise fuzzy memberships to the classification target space. The linear coefficients are the rule's output constants $y_{j,c}$ in TSK terminology.

In the proposed layer the input *fuzzification* step (embed : $\mathbb{R} \to [0,1]^d$) plays the dual role: a single linear map $\mathbb{R} \to \mathbb{R}^d$ followed by a σ produces d memberships from the crisp scalar input. This is the only embedding-side affine, again TSK-justified.

6 Linguistic hedges and learnable gates

6.1 Zadeh's hedges as power operators

Definition 6.1 (Linguistic hedge, after Zadeh [2])**.** *A* linguistic hedge *is a unary operator* $h : [0,1] \to [0,1]$ *that modifies a membership function to express a linguistic refinement. Canonical hedges are:*

- ***Concentration*** *(*"very"*):* $h_{\mathrm{con}}(\mu) = \mu^2$ *(or* μ^p *for* $p > 1$*).*
- ***Dilation*** *(*"more or less"*):* $h_{\mathrm{dil}}(\mu) = \sqrt{\mu}$ *(or* $\mu^{1/p}$*).*
- ***Intensification****: piecewise definition* $h_{\mathrm{int}}(\mu) = 2\mu^2$ *for* $\mu \le \frac{1}{2}$, $1 - 2(1-\mu)^2$ *otherwise — a smooth S-curve fixing* $0, \frac{1}{2}, 1$*.*

The exponent-form hedges form a one-parameter family $\mu \mapsto \mu^p$ with $p \in (0,\infty)$, monotonically interpolating between dilation ($p < 1$), identity ($p = 1$), and concentration ($p > 1$). With p promoted to a learnable scalar, gradient descent can shape the membership exactly to the task.

6.2 Sigmoidal hedges and the dilation parameter τ

A second one-parameter family arises from the sigmoid:

$$h_\tau(x) \;=\; \sigma(\tau(x-c)), \qquad \tau > 0,\;\; c \in [0,1] \;\; \text{(fixed centre)}. \tag{10}$$

- As $\tau \to 0^+$, $h_\tau(x) \to \frac{1}{2}$ uniformly: a degenerate hedge that erases all sharpness.
- As $\tau \to +\infty$, $h_\tau \to \mathbb{1}_{[c,1]}$: the crisp Heaviside; the hedge "cuts" the universe at c.
- For intermediate τ, h_τ is a smooth approximation of the step at c, with steepness $\tau/4$ at $x = c$.

The construction is the smooth analog of a thresholding operator and is the natural shape for an *Atanassov mixer gate* (Section 3). With $c = \frac{1}{2}$ and τ a learnable scalar per channel, the gate

$$g_\tau(x) \;=\; \sigma\big(\tau\,(x - \tfrac{1}{2})\big)$$

implements a per-channel learnable hedge: each channel can independently choose how sharply membership tips from μ^- to μ^+.

6.3 Boundary behaviour of the learnable gate

Lemma 6.1 ([0, 1]-preservation of the Atanassov mix)**.** *Let $\mu^+, \mu^- \in [0,1]$ and $g \in [0,1]$. Then $g\,\mu^+ + (1-g)\,\mu^- \in [0,1]$ and $g\,\mu^+ + (1-g)\,\mu^-$ depends monotonically and continuously on each of μ^+, μ^-, g.*

Proof. Convex combination of points in $[0,1]$. Continuity and monotonicity are elementary. □

Lemma 6.2 (Gradient regularity of g_τ)**.** *For any fixed $x \in [0,1]$ and $c \in [0,1]$, the map $\tau \mapsto g_\tau(x)$ is C^∞, with*

$$\partial_\tau\, g_\tau(x) \;=\; (x-c)\, g_\tau(x)\big(1 - g_\tau(x)\big),$$

which vanishes only at $x = c$ or in the saturation limits $\tau \to \pm\infty$.

Proof. Direct differentiation of (10); the chain rule and $\sigma'(u) = \sigma(u)(1-\sigma(u))$. □

Lemma 6.2 ensures that gradient updates on τ are well-conditioned except at the trivial $x = c$ (where the gate has no preference) and at extreme saturation, exactly the regimes where one would not want the gate to move anyway.

6.4 OWA and ordered weighted hedges

Definition 6.2 (Ordered Weighted Aggregation, after Yager [8])**.** *Given weights $\mathbf{w} = (w_1, \ldots, w_n) \in [0,1]^n$ with $\sum_i w_i = 1$ and inputs $\mathbf{a} = (a_1, \ldots, a_n) \in [0,1]^n$,*

$$\mathrm{OWA}_{\mathbf{w}}(\mathbf{a}) \;=\; \sum_{i=1}^{n} w_i\, a_{(i)},$$

where $a_{(1)} \geq a_{(2)} \geq \cdots \geq a_{(n)}$ is the sorted input.

OWA is a parameterised aggregator that interpolates the canonical operators:

- $\mathbf{w} = (1, 0, \ldots, 0) \Rightarrow$ OWA = max (Gödel t-conorm S_G).
- $\mathbf{w} = (0, \ldots, 0, 1) \Rightarrow$ OWA = min (Gödel t-norm T_G).
- $\mathbf{w}_i = 1/n \Rightarrow$ OWA = mean (arithmetic average).

The HSiKAN multi-arity mixer (Remark 4.1) is a constrained OWA: the weights $\alpha_k = \mathrm{softmax}(\boldsymbol{\ell})_k$ are softmax-normalised and applied to a small set of arity branches (typically $|\mathcal{K}| \in \{2, 3\}$). It is not a per-position sort-based OWA but a per-scale weighted aggregator with the same constraints.

7 The HSiKAN–Kóczy correspondence

We can now state the correspondence between HSiKAN's existing mechanisms and Kóczy-fuzzy-signature primitives in exhaustive form.

7.1 Catmull–Rom spline as learnable membership function

Definition 7.1 (Catmull–Rom interpolant, after [12])**.** *For control points $p_0, p_1, p_2, p_3 \in \mathbb{R}$ at abscissae $x_{-1} < x_0 < x_1 < x_2$ with uniform spacing, the Catmull–Rom interpolant on $[x_0, x_1]$ at fractional position $t \in [0,1]$ is*

$$\mathrm{CR}(t; p_0, p_1, p_2, p_3) = \frac{1}{2}\Big[2p_1 + (-p_0 + p_2)t + (2p_0 - 5p_1 + 4p_2 - p_3)t^2 + (-p_0 + 3p_1 - 3p_2 + p_3)t^3\Big].$$

A full Catmull–Rom curve over m control points $\mathbf{p} = (p_0, \ldots, p_{m-1})$ is the piecewise concatenation of these cubic patches. The curve is C^1 at every interior knot and exactly interpolates $p_1, \ldots, p_{m-2}$.

As a learnable membership function. With m control points on a fixed grid of $X = \mathbb{R}$ (in the proposed layer, $X = [-3, 3]$ as input domain after a fuzzification step), and with the output post-processed by $\sigma\colon \mathbb{R} \to [0,1]$, the Catmull–Rom curve realises a learnable membership function $\mu_\theta\colon \mathbb{R} \to [0,1]$ parameterised by $\theta = \mathbf{p} \in \mathbb{R}^m$. The parameter count is m per channel per branch.

The Catmull–Rom basis is preferred over a generic spline basis for two reasons: (i) the interpolant passes through its control points exactly (at the interior knots), so the parameters *are* the membership values at the grid, easing interpretability; (ii) the local support of each control point (4 neighbouring segments) means a gradient at a single sample updates at most 4 control points, keeping the layer's effective capacity modest.

7.2 The full correspondence table (formal)

The table on the next page formally pairs every learnable degree of freedom of the proposed layer with a named fuzzy-system primitive.

HSiKAN piece (existing)	Fuzzy-systems analog (named)
`embed`: $\mathbb{R} \to \mathbb{R}^d + \sigma$	Fuzzification rule (TSK input fuzzifier). Maps a crisp value to d fuzzy memberships in $[0,1]^d$. Sugeno-style.
$\texttt{CR}^+$ (branch 0): m control points, post-σ	Learnable membership $\mu^+ : [0,1] \to [0,1]$ of the Atanassov pair.
$\texttt{CR}^-$ (branch 1): m control points, post-σ	Learnable non-membership μ^- of the Atanassov pair. Together with μ^+ forms the IFS (μ, ν) after the gate.
Polarity gate $g(x)$, learnable scalar τ	Sigmoidal linguistic hedge $\sigma(\tau(x-c))$ (Definition 6.1). One τ per channel per layer.
$v \to e$ pooling T (categorical)	T-norm conjunction over hyperedge members (Section 2).
$e \to v$ pooling S (categorical)	T-conorm disjunction over edges incident to a vertex. Replaces the prior weighted-sum scatter to keep $[0,1]$ end-to-end.
W_e per-edge weight	Rule strength / edge credibility weight. Multiplies μ_e before the t-conorm step. Can be tied ($W_e =$ scalar) for equivariance, or per-edge to permit "rule salience" learning.
W_{pos} per-position weight (optional)	Position-conditional membership factor inside the hyperedge. Constrained to $[0,1]$ via $\sigma(\boldsymbol{\ell})$ when active.
α_k softmax over arities (optional)	OWA across granularities (Remark 4.1).
Global mean over V + linear head	Sugeno defuzzification + TSK output rule (Section 5).

7.3 Parameter accounting

For the default configuration (d, L, $m = 8$, $n_{\text{cls}} = 10$, no W_{pos}, no multi-arity, W_e tied) the parameter count decomposes as

$$N_{\text{params}}(d, L) \;=\; \underbrace{2d}_{\text{embed}} \;+ L\,\underbrace{\left(2dm \;+\; d \;+\; 1\right)}_{\text{per-layer}} \;+\; \underbrace{(d+1)\,n_{\text{cls}}}_{\text{head}}.$$

The per-layer term decomposes as: $2dm$ for the two CR branches of the Atanassov pair, d for the per-channel τ scalars, and 1 for the tied W_e scalar (which represents the single rule strength of the layer).

At $d = 16$, $L = 8$:

$$N_{\text{params}}(16, 8) \;=\; 32 + 8\,(2 \cdot 16 \cdot 8 + 16 + 1) + 17 \cdot 10 \;=\; 32 + 8 \cdot 273 + 170 \;=\; 2{,}386.$$

8 The proposed FuzzySignatureLayer in fuzzy-mathematical terms

We now state the proposed layer as a composition of named fuzzy-system operators, with all preconditions, postconditions, and $[0,1]$-boundary guarantees explicit.

8.1 Notation

- Image $\mathbf{I} \in [0,1]^{H \times W}$ (grayscale, fuzzified by `ToTensor`).
- Vertices $V = \{(i,j) : 0 \le i < H,\, 0 \le j < W\}$, $|V| = HW$.

- Hyperedges E indexed by receptive-field anchor positions, each edge containing the $K = k^2$ vertices of a $k \times k$ patch.
- Edge-members table $\mathrm{em} \in V^{|E| \times K}$ (long indices).
- Per-vertex incident-edges table $\mathrm{ve} \in E^{|V| \times M}$ with mask $\mathrm{m} \in \{0,1\}^{|V| \times M}$, where M is the maximum number of edges any vertex belongs to.
- Layer dimension d, layer count L, control-point count m.

8.2 Layer forward

Definition 8.1 (FuzzySignatureLayer forward)**.** *Input* $\mathbf{x} \in [0,1]^{|V| \times d}$ *(vector-valued fuzzy memberships per vertex per channel). The layer parameters are* $(\mathbf{P}^+, \mathbf{P}^-, \boldsymbol{\tau}, W_e)$ *with* $\mathbf{P}^\pm \in \mathbb{R}^{d \times m}$, $\boldsymbol{\tau} \in \mathbb{R}^d_{>0}$, $W_e \in \mathbb{R}$ *(or* $\mathbb{R}^{|E|}$ *if untied). The forward is the following composition:*

$$
\begin{aligned}
\mu_v^+ &= \sigma\big(\mathrm{CR}(\mathbf{x}_v; \mathbf{P}^+)\big) && \in [0,1]^{|V| \times d} && \text{(F1)}\\
\mu_v^- &= \sigma\big(\mathrm{CR}(\mathbf{x}_v; \mathbf{P}^-)\big) && \in [0,1]^{|V| \times d} && \text{(F2)}\\
g_v &= \sigma\big(\boldsymbol{\tau} \odot (\mathbf{x}_v - \tfrac{1}{2})\big) && \in [0,1]^{|V| \times d} && \text{(F3)}\\
\mu_v &= g_v \odot \mu_v^+ + (1 - g_v) \odot \mu_v^- && \in [0,1]^{|V| \times d} && \text{(F4)}\\
\mathbf{h}_e &= \mathsf{T}\big(\{\mu_v : v \in e\}\big) && \in [0,1]^{|E| \times d} && \text{(F5)}\\
\tilde{\mathbf{h}}_e &= \min\big(W_e,\, 1\big) \odot \mathbf{h}_e && \in [0,1]^{|E| \times d} && \text{(F6)}\\
\mathbf{h}_v &= \mathsf{S}\big(\{\tilde{\mathbf{h}}_e : e \ni v\}\big) && \in [0,1]^{|V| \times d} && \text{(F7)}\\
\mathbf{x}_v^{\mathrm{out}} &= \tfrac{1}{2}\big(\mathbf{x}_v + \mathbf{h}_v\big) && \in [0,1]^{|V| \times d} && \text{(F8)}
\end{aligned}
$$

where $\odot$ *is element-wise product,* σ *is the logistic function,* T, S *are the categorical t-norm and t-conorm of the layer, and* $\min(W_e, 1)$ *projects* W_e *into* $[0,1]$ *(a Lipschitz projection that preserves gradient flow for* $W_e \le 1$*).*

Theorem 8.1 ($[0,1]$-preservation through arbitrary depth)**.** *If* $\mathbf{x} \in [0,1]^{|V| \times d}$ *then* $\mathbf{x}^{\mathrm{out}} \in [0,1]^{|V| \times d}$. *Composing* L *such layers preserves* $[0,1]$ *regardless of* L.

Proof. Equations (F1)–(F3) are σ outputs and lie in $[0,1]$ by definition of the logistic function. Equation (F4) is a convex combination of μ_v^+ and μ_v^- governed by $g_v \in [0,1]$ — Lemma 6.1. Equation (F5) applies a t-norm to a tuple in $[0,1]^K$ and lies in $[0,1]$ by Definition 2.2. Equation (F6) multiplies by $\min(W_e, 1) \in [0,1]$ when $W_e \ge 0$; when $W_e < 0$, the layer takes the absolute value (a fuzzy negation has been applied to the rule strength) — clamp to $[0,1]$ for stability. Equation (F7) applies a t-conorm to a tuple in $[0,1]^M$ with mask and lies in $[0,1]$ by Definition 2.3. Equation (F8) is a convex combination $\frac{1}{2}\mathbf{x}_v + \frac{1}{2}\mathbf{h}_v$ of two elements of $[0,1]$, so it is in $[0,1]$. Composition preserves $[0,1]$ since each layer is a function $[0,1]^{|V| \times d} \to [0,1]^{|V| \times d}$. □

Theorem 8.2 (Monotonicity in inputs)**.** *Holding* $(\mathbf{P}^+, \mathbf{P}^-, \boldsymbol{\tau}, W_e)$ *fixed and* assuming $\mathbf{P}^+, \mathbf{P}^-$ *such that the* $\sigma \circ \mathrm{CR}$ *outputs are monotone in their argument (which holds when the control points are monotone in* θ*, true at init under* $0.05 \cdot \mathcal{N}$ *random initialisation in expectation), the layer forward is monotone in each* $\mathbf{x}_v$*: increasing one component does not decrease* $\mathbf{x}_v^{\mathrm{out}}$.

Proof sketch. Equations (F1)–(F3) are componentwise monotone in their arguments by monotonicity of the spline and of σ. Equation (F4) preserves componentwise monotonicity provided $\mu_v^+ \ge \mu_v^-$, which is the inductive bias of the layer (the positive branch carries the positive membership). This holds at initialisation in expectation, but is not a hard invariant — the layer is not strictly monotone late in training. The remaining equations preserve monotonicity by axioms of t-norms and t-conorms. □

8.3 Classifier wrapping

Definition 8.2 (FuzzySignatureClassifier). *For inputs $\mathbf{I} \in [0,1]^{1\times H\times W}$ and n_{cls} classes, the classifier is the composition*

$$\begin{aligned}
\mathbf{x}_v^{(0)} &= \sigma\big(W_{\text{emb}}\,\mathbf{I}_v + \mathbf{b}_{\text{emb}}\big) && \in [0,1]^{|V|\times d} \quad \textit{(fuzzification)} && \text{(C1)}\\
\mathbf{x}^{(\ell)} &= \text{FuzzySignatureLayer}^{(\ell)}\big(\mathbf{x}^{(\ell-1)}\big) && \ell = 1,\dots,L && \text{(C2)}\\
\mathbf{z} &= \tfrac{1}{|V|}\sum_v \mathbf{x}_v^{(L)} && \in [0,1]^d \quad \textit{(Sugeno pool)} && \text{(C3)}\\
\boldsymbol{\ell} &= W_{\text{head}}\,\mathbf{z} + \mathbf{b}_{\text{head}} && \in \mathbb{R}^{n_{\text{cls}}} \quad \textit{(TSK output)} && \text{(C4)}
\end{aligned}$$

The two linear maps that survive ((C1), (C4)) are the *fuzzification* and *TSK output rule* prescribed by Sugeno's defuzzification (Section 5). They are not generic `Linear` layers but identified primitives of the TSK fuzzy system architecture.

8.4 What is *not* in the layer

We list, for clarity, the operators *omitted* from the proposed design, with justification:

- No internal `Linear` layers between fuzzification and defuzzification: would constrain the layer's plasticity into the affine cone, contradicting the user-mandated design principle that HSiKAN's plasticity should not be reduced.
- No internal sigmoid layers beyond those already justified ($\sigma \circ \mathrm{CR}$ for membership clamping; $\sigma(\tau(x-c))$ for the hedge gate; σ at the fuzzification step). The only sigmoids in the layer are *identified* fuzzy operators.
- No layernorm or batchnorm: would break the $[0,1]$-boundary and the fuzzy interpretation.
- No dropout: incompatible with deterministic fuzzy-system interpretation. (At training time, a fuzzy dropout could be introduced as a multiplicative hedge — left as future work.)

8.5 Inductive biases captured

The layer encodes the following inductive biases by construction:

1. **Locality**: hyperedges are receptive-field patches; information aggregates within $k\times k$ neighbourhoods.
2. **Boundedness**: all internal signals are in $[0,1]$; the network cannot produce unbounded activations.
3. **Atanassov duality**: each channel learns a (μ^+,μ^-) pair with a per-channel hedge τ; the layer is intuitionistic by construction.
4. **Sugeno output**: the layer head is a TSK output rule, not a generic projection.
5. **Compositionality**: L layers are L depth-1 signatures, but Theorem 8.1 guarantees compositional safety to arbitrary depth without normalisation.

9 Practical observations and pitfalls

9.1 Łukasiewicz t-norm saturation at large arity

The Łukasiewicz t-norm $\mathsf{T}_{\mathrm{L}}(a_1,\dots,a_n) = \max(0, \sum_i a_i - (n-1))$ saturates to 0 at $n = K = 25$ (kernel $k=5$) for inputs $a_i \sim \mathrm{Uniform}(0,1)$ in expectation: the threshold $K-1 = 24$ vs. expected sum $K/2 = 12.5$ leaves the t-norm at the saturation boundary almost surely at initialisation. Practical consequence: T_{L} is a known-poor t-norm for the RF-size regime considered here. It is included for completeness and as a worst-case reference; the design does not depend on it for empirical performance.

9.2 Product t-norm gradient on long products

For product t-norm with K factors, $\mathsf{T}_\mathrm{P}(a_1, \ldots, a_K) = \prod_i a_i$. At inputs near 0.5, $0.5^{25} \approx 3 \times 10^{-8}$, which underflows in single precision. The implementation uses the log-sum form $\exp(\sum_i \log a_i)$ with a_i clamped to $[\epsilon, 1]$ for stability. The gradient $\partial \mathsf{T}_\mathrm{P}/\partial a_j = \prod_{i \neq j} a_i$ is similarly small at $0.5^{24} \approx 6 \times 10^{-8}$; this is the expected behaviour of the product t-norm and motivates the use of T_G (which has a cleaner gradient) as the default for the proposed layer in the empirical sweeps.

9.3 Sigmoid saturation at the CR-membership output

If $|\mathrm{CR}(x; \mathbf{P})| \gg 3$ the post-σ output saturates and gradient vanishes. The initialisation $\mathbf{P} \sim 0.05 \cdot \mathcal{N}(0, 1)$ keeps CR small at init, so σ operates near $\frac{1}{2}$ where gradient is maximal. Late in training, saturation is acceptable: it indicates the membership has committed to a region.

9.4 Learnable τ: positivity constraint

The hedge dilation τ must be positive to preserve the hedge interpretation (negative τ would invert the direction). Two common parametrisations: (i) softplus, $\tau = \log(1 + e^\ell)$, with unconstrained ℓ; (ii) exponential, $\tau = e^\ell$. The layer uses softplus to avoid blow-up at large ℓ and to keep $\tau \to 0$ attainable.

10 Worked examples

This section illustrates the operators of Sections 2–6 on small numerical inputs, sufficient to verify each implementation by hand-calculation and to expose the orderings claimed in Propositions 2.1–2.3.

10.1 T-norms at $K = 4$

Let $\mathbf{a} = (0.8, 0.6, 0.9, 0.7) \in [0, 1]^4$.

$$\mathsf{T}_\mathrm{G}(\mathbf{a}) = \min(0.8, 0.6, 0.9, 0.7) = 0.6, \tag{11}$$
$$\mathsf{T}_\mathrm{P}(\mathbf{a}) = 0.8 \cdot 0.6 \cdot 0.9 \cdot 0.7 = 0.3024, \tag{12}$$
$$\mathsf{T}_\mathrm{L}(\mathbf{a}) = \max(0,\ 0.8 + 0.6 + 0.9 + 0.7 - 3) = \max(0, 0.0) = 0.0. \tag{13}$$

The ordering $\mathsf{T}_\mathrm{L} = 0.0 \leq \mathsf{T}_\mathrm{P} = 0.3024 \leq \mathsf{T}_\mathrm{G} = 0.6$ holds, in agreement with Proposition 2.1.

10.2 T-norms at $K = 25$ (kernel size 5)

Let all $a_i = 0.5$ for $i = 1, \ldots, 25$:

$$\mathsf{T}_\mathrm{G} = 0.5, \tag{14}$$
$$\mathsf{T}_\mathrm{P} = 0.5^{25} \approx 2.98 \times 10^{-8}, \tag{15}$$
$$\mathsf{T}_\mathrm{L} = \max(0,\ 12.5 - 24) = 0. \tag{16}$$

At RF size 5×5, T_P underflows in single precision without the log-sum reformulation, and T_L saturates dead to zero at typical fuzzy inputs 0.5. Both observations motivate the default T_G choice in the proposed layer's smoke run.

10.3 T-conorms at $K = 4$

Same inputs $\mathbf{a}$ as above:

$$\mathsf{S}_{\mathrm{G}}(\mathbf{a}) = \max(\mathbf{a}) = 0.9, \tag{17}$$
$$\mathsf{S}_{\mathrm{P}}(\mathbf{a}) = 1 - (1-0.8)(1-0.6)(1-0.9)(1-0.7) = 1 - 0.2 \cdot 0.4 \cdot 0.1 \cdot 0.3 = 1 - 0.0024 = 0.9976, \tag{18}$$
$$\mathsf{S}_{\mathrm{L}}(\mathbf{a}) = \min(1,\ 0.8 + 0.6 + 0.9 + 0.7) = 1.0. \tag{19}$$

The ordering $0.9 \le 0.9976 \le 1.0$ confirms Proposition 2.3.

10.4 Parametric Hamacher t-norm at $\gamma \in \{0, 1, 2, 5\}$

For $a = 0.8, b = 0.6$:

$$\gamma = 0: \quad \frac{ab}{0 + 1 \cdot (a + b - ab)} = \frac{0.48}{0.92} \approx 0.522, \tag{20}$$
$$\gamma = 1: \quad \frac{0.48}{1 + 0} = 0.48 = \mathsf{T}_{\mathrm{P}}(0.8, 0.6), \tag{21}$$
$$\gamma = 2: \quad \frac{0.48}{2 - 0.92} = \frac{0.48}{1.08} \approx 0.444, \tag{22}$$
$$\gamma = 5: \quad \frac{0.48}{5 - 4 \cdot 0.92} = \frac{0.48}{1.32} \approx 0.364. \tag{23}$$

As γ increases from 0, the Hamacher family monotonically descends from a t-norm above T_{P} towards a Drastic-Product–like limit. At $\gamma = 1$ it coincides exactly with T_{P}, the intended interpretation of the algebraic t-norm as the Hamacher-canonical reference point.

10.5 Yager t-norm for $p \in \{1, 2, \infty\}$

For $a = 0.8, b = 0.6$:

$$p = 1: \quad 1 - \min(1, (0.2 + 0.4)) = 1 - 0.6 = 0.4 = \mathsf{T}_{\mathrm{L}}(0.8, 0.6), \tag{24}$$
$$p = 2: \quad 1 - \min(1, \sqrt{0.04 + 0.16}) = 1 - \sqrt{0.2} \approx 0.553, \tag{25}$$
$$p \to \infty: \quad 1 - \max(0.2, 0.4) = 1 - 0.4 = 0.6 = \mathsf{T}_{\mathrm{G}}(0.8, 0.6). \tag{26}$$

Yager's parameter p interpolates from T_{L} ($p = 1$) through T_{P}-like values towards T_{G} ($p \to \infty$), spanning the entire canonical ordering.

10.6 Aczél–Alsina t-norm for $\lambda \in \{1, 2, 5\}$

Same $a = 0.8, b = 0.6$, with $-\log 0.8 \approx 0.223$, $-\log 0.6 \approx 0.511$:

$$\lambda = 1: \quad \exp(-(0.223 + 0.511)) = \exp(-0.734) \approx 0.48 = \mathsf{T}_{\mathrm{P}}, \tag{27}$$
$$\lambda = 2: \quad \exp(-\sqrt{0.0497 + 0.261}) = \exp(-0.557) \approx 0.573, \tag{28}$$
$$\lambda = 5: \quad \exp(-(0.223^5 + 0.511^5)^{1/5}) \approx \exp(-0.513) \approx 0.599. \tag{29}$$

Aczél–Alsina with $\lambda = 1$ reproduces T_{P}; as $\lambda \to \infty$ it converges to T_{G}. The single λ parameter therefore covers exactly the canonical interval $[\mathsf{T}_{\mathrm{P}}, \mathsf{T}_{\mathrm{G}}]$, complementing Yager's $[\mathsf{T}_{\mathrm{L}}, \mathsf{T}_{\mathrm{G}}]$ range.

10.7 Atanassov-pair example

Suppose at a vertex v we have crisp input $x_v = 0.7$, and the trained CR branches evaluate to $\mu^+(0.7) = 0.85$, $\mu^-(0.7) = 0.25$. With learnable $\tau = 6.0$, $c = 0.5$:

$$
\begin{aligned}
g_\tau(0.7) &= \sigma(6.0 \cdot (0.7 - 0.5)) = \sigma(1.2) \approx 0.769, && (30)\\
\mu(0.7) &= 0.769 \cdot 0.85 \approx 0.654, && (31)\\
\nu(0.7) &= (1 - 0.769) \cdot 0.25 = 0.231 \cdot 0.25 \approx 0.058, && (32)\\
\pi(0.7) &= 1 - 0.654 - 0.058 \approx 0.288. && (33)
\end{aligned}
$$

The IFS constraint $\mu + \nu \leq 1$ holds (with ample slack). The hesitancy $\pi \approx 0.288$ quantifies the layer's uncertainty about the membership of this vertex: even after the gate, almost 30% of the unit interval is undetermined.

10.8 Effect of varying τ

For fixed CR outputs $\mu^+(x) = 0.85$, $\mu^-(x) = 0.25$, $x = 0.7$, $c = 0.5$:

τ	g_τ	$\mu + \nu$	π
0.5	0.525	$0.525 \cdot 0.85 + 0.475 \cdot 0.25 = 0.565$	0.435
2.0	0.599	$0.599 \cdot 0.85 + 0.401 \cdot 0.25 = 0.610$	0.390
6.0	0.769	$0.769 \cdot 0.85 + 0.231 \cdot 0.25 = 0.712$	0.288
20.0	0.982	$0.982 \cdot 0.85 + 0.018 \cdot 0.25 = 0.839$	0.161
∞	1.0	$1 \cdot 0.85 + 0 \cdot 0.25 = 0.850$	0.150

As τ increases, the gate sharpens towards a crisp commitment to μ^+ (since $x > c$), the IFS support $\mu + \nu$ rises, and the hesitancy falls towards $1 - \mu^+ = 0.15$. Conversely small τ keeps the layer indecisive. The learnable τ lets gradient descent calibrate sharpness per channel from the data.

11 Derivation of the TSK output rule

We give a self-contained derivation of why the classifier head (Equation (C4)) is the canonical Takagi–Sugeno–Kang *output rule*, not a generic linear map.

11.1 The TSK fuzzy system

A Takagi–Sugeno–Kang (TSK) zero-order fuzzy system [11] consists of R rules of the form

$$\mathrm{R}_r: \quad \text{IF } x_1 \text{ is } A_{r,1} \text{ AND } \cdots \text{ AND } x_p \text{ is } A_{r,p} \quad \text{THEN } y = y_r,$$

where $A_{r,j}$ are fuzzy sets and $y_r \in \mathbb{R}$ is the rule's output constant. The firing strength of rule r at crisp input $\mathbf{x} = (x_1, \ldots, x_p)$ is

$$w_r(\mathbf{x}) = \mathsf{T}\big(\mu_{A_{r,1}}(x_1), \ldots, \mu_{A_{r,p}}(x_p)\big).$$

The system's output is the weighted-average defuzzification:

$$y(\mathbf{x}) = \frac{\sum_{r=1}^{R} w_r(\mathbf{x})\, y_r}{\sum_{r=1}^{R} w_r(\mathbf{x})}.$$

11.2 TSK as a neural architecture

If we identify

- the input vertices $v \in V$ with TSK rules (one rule per vertex, with $V = HW$ rules per layer),
- the per-channel membership $\mu_v[c]$ with the firing strength of rule v on channel c,
- the linear-head row $W_{\text{head}}[k,:] \in \mathbb{R}^d$ with the per-channel output constants $\{y_{r,k}\}_{c=1}^{d}$ for class k,

then the classifier head's output for class k is

$$\ell_k = \sum_{v \in V} \sum_{c=1}^{d} \frac{1}{|V|}\, \mu_v[c]\, W_{\text{head}}[k,c] + b_k.$$

Here the normalisation $1/|V|$ replaces the usual $1/\sum_r w_r$ — a simplification valid because Theorem 8.1 bounds each $\mu_v[c] \in [0,1]$ so $\sum_v \mu_v[c] \leq |V|$. The denominator $|V|$ is a fixed Lipschitz upper bound; the bias b_k absorbs any residual offset.

Thus the linear head is the TSK output rule, with channels playing the role of rules. The classifier is a $V \cdot d$-rule TSK system whose output constants are jointly learnt with the input fuzzifier (embed Linear) and the rule firing strengths (the layer parameters $\mathbf{P}^{\pm}, \boldsymbol{\tau}, W_e$).

11.3 Sugeno integral vs. weighted average

The Sugeno integral

$$\mathrm{Sug}_w(\mu) = \sup_{t \in [0,1]} \big[t \wedge w(\{x : \mu(x) \geq t\})\big]$$

is the non-additive cousin of the weighted average, replacing the sum with sup and product with $\wedge$. For neural training the non-differentiable sup is undesirable, and the weighted average is preferred — this is the established choice in the TSK family rather than in Mamdani-style systems. The proposed layer follows the TSK convention.

12 OWA, generalised means, and the multi-arity mixer

12.1 Generalised mean as one-parameter aggregator

The *generalised mean* of $\mathbf{a} \in [0,1]^n$ with parameter $r \in \mathbb{R}$ is

$$M_r(\mathbf{a}) = \Big(\tfrac{1}{n}\sum_{i=1}^{n} a_i^r\Big)^{1/r}, \qquad r \neq 0; \qquad M_0(\mathbf{a}) = \Big(\prod_i a_i\Big)^{1/n}.$$

- $r \to -\infty$: $M_r \to \min$.
- $r = -1$: harmonic mean.
- $r = 0$: geometric mean (continuous extension).
- $r = 1$: arithmetic mean.
- $r = 2$: quadratic mean.
- $r \to +\infty$: $M_r \to \max$.

The generalised mean is monotonic in r (for fixed $\mathbf{a}$, M_r is non-decreasing in r). With r a learnable scalar (constrained positive via softplus, or signed via no constraint), gradient descent chooses where on the min–max axis the aggregator sits.

12.2 OWA vs. generalised mean

OWA and the generalised mean are distinct parametric aggregator families: the OWA depends only on the multiset of inputs (via sorting), while the generalised mean depends on the unordered inputs but inherits the min-to-max interpolation. OWA admits any $[0, 1]$-valued weight family that sums to 1 (so its dimension is $n - 1$ free parameters), whereas M_r has a single scalar. The proposed layer uses M_r-style adjacent t-norms (Hamacher γ, Yager p) only as a future extension; the multi-arity mixer is a $|\mathcal{K}|$-OWA over RF scales.

12.3 Worked example: OWA across three scales

Suppose three arity branches return per-vertex memberships $\{\mu_v^{(k)}\}_{k=1}^{3} = \{0.4, 0.7, 0.2\}$ at vertex v.

- Softmax of equal logits $(0, 0, 0)$: $\alpha = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, OWA $= (0.4+0.7+0.2)/3 \approx 0.433$ (arithmetic mean).
- Softmax of $(0, 2, 0)$: $\alpha \approx (0.106, 0.788, 0.106)$, OWA $\approx 0.106 \cdot 0.4 + 0.788 \cdot 0.7 + 0.106 \cdot 0.2 \approx 0.616$, biased towards the second branch.
- Softmax of $(0, 6, 0)$: $\alpha \approx (0.001, 0.998, 0.001)$, OWA ≈ 0.7, effectively the second branch only.

As the softmax sharpens, the multi-arity mixer transitions from a uniform mean across scales to a selection of one dominant scale. The gradient on $\boldsymbol{\ell}$ continuously moves the layer along this axis, with the same one-parameter-family flavour as the τ hedge.

13 Connection to neuro-fuzzy literature

The proposed layer is one specific realisation of the wider *neuro-fuzzy* programme [13, 14]. We locate it within that programme:

- **ANFIS** [13] is a TSK system with Gaussian-bell membership functions per input and a single weighted-average output. The proposed layer differs by (a) learnable Catmull–Rom memberships in place of parametric Gaussians, (b) hierarchical hypergraph propagation in place of single-layer rules, (c) Atanassov pair instead of single membership, (d) categorical t-norm/t-conorm choice instead of product/sum default.
- **FALCON** [15] and **NEFCLASS** [16] use neural-style learnability with classical Mamdani fuzzy systems. The proposed layer is closer to a *convolutional* TSK system than to a Mamdani classifier.
- **Kóczy fuzzy signatures** [5, 6] are the closest direct predecessor. The proposed layer instantiates a depth-L fuzzy signature where the tree structure is the receptive-field hypergraph, the aggregators are categorical at the layer level, and every other parameter is learnable.

The added value of the proposed layer over its neuro-fuzzy predecessors is the *hypergraph propagation* step: each vertex aggregates a multiplicity of overlapping hyperedges via the t-conorm (Equation F7), giving each pixel access to multiple receptive-field contexts at every depth. This is the "hypergraph signed convolution" contribution; the rest is classical.

14 Open mathematical questions

We list four open questions that the proposed design surfaces but does not answer:

1. **Optimal categorical aggregator for vision regime.** Section 10 shows that T_{L} saturates at $K = 25$, and T_{P} underflows. Is T_{G} the unique stable choice at this RF size, or do parametric families (γ, p, λ) open intermediate operating points?

2. **Theoretical capacity of CR-as-membership.** How does a Catmull–Rom interpolant with $m = 8$ control points compare, in approximation power on the class of bounded Lipschitz membership functions, to a Gaussian with two parameters or a generalised bell with three? A piecewise universal approximation bound for the C^1 Catmull–Rom basis on $[0, 1]$ would tighten the parameter-budget argument.
3. **IFS aggregation theorems for the layer.** Section 3 states two natural IFS aggregators (T- and T^c-aggregate). Equation (F5)–(F7) implement the T-aggregate on μ but lose the ν track entirely: μ^- is consumed inside Equation (F4). A faithful IFS layer would propagate (μ, ν) as a *pair* of tensors with two t-norms; this doubles the memory cost. Whether the carried-along ν buys empirical accuracy is an open empirical question.
4. **Learnable categorical aggregators.** The t-norm and t-conorm are presently categorical hyperparameters. A continuous relaxation — interpolating $\mathsf{T}_\mathrm{G}, \mathsf{T}_\mathrm{P}, \mathsf{T}_\mathrm{L}$ by a softmax-weighted convex combination, with weights learnable per-layer — would let gradient descent choose the t-norm. Whether such a relaxation maintains t-norm axioms (associativity especially) for non-trivial mixing weights is open; the answer is *no* in general, but *yes* for certain restricted parametric families (Hamacher, Yager).

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